

Bridge Structural Health Monitoring Using Graph Neural Networks

Deep Learning Project · GCN-Based SHM for Steel Truss Bridges





Problem Statement & Objectives

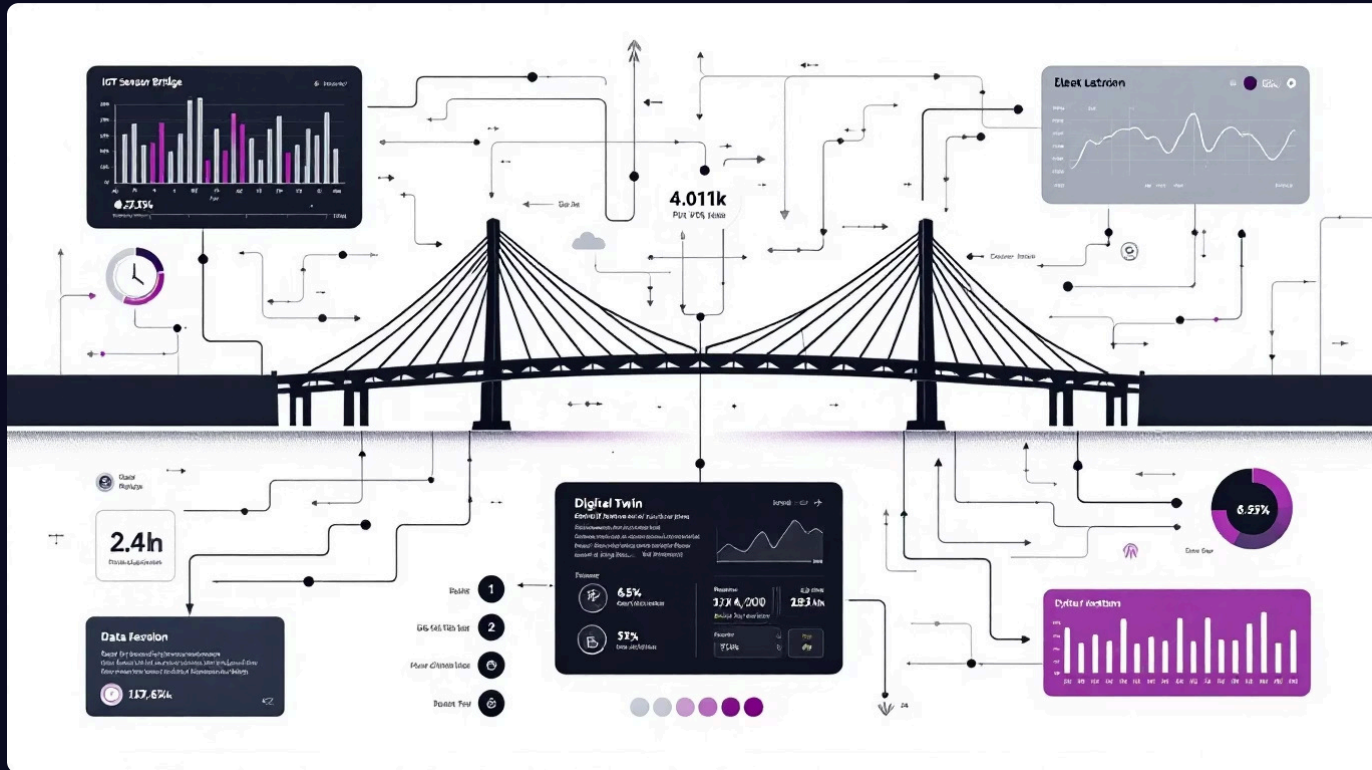
The Problem

- Traditional SHM relies on periodic inspections — damage accumulates undetected between visits
- Sensor channels treated as independent feature vectors
- Physical relationships between structural domains are ignored

Our Objectives

- Capture inter-domain relationships using graph-based learning
- Improve prediction accuracy over conventional baselines
- Compare GCN against a 1D CNN under identical data conditions
- Build an end-to-end pipeline with explainability and maintenance output

Dataset Overview



The primary dataset is a **36 MB digital twin simulation** of a smart bridge, comprising minute-by-minute IoT sensor readings. A secondary synthetic dataset enables cross-domain generalisation experiments.

Binary Label: Maintenance_Alert — 0 = Healthy, 1 = Requires Intervention

50,000+ Rows

Real-world digital twin records

25 Features

Across 5 sensor domains

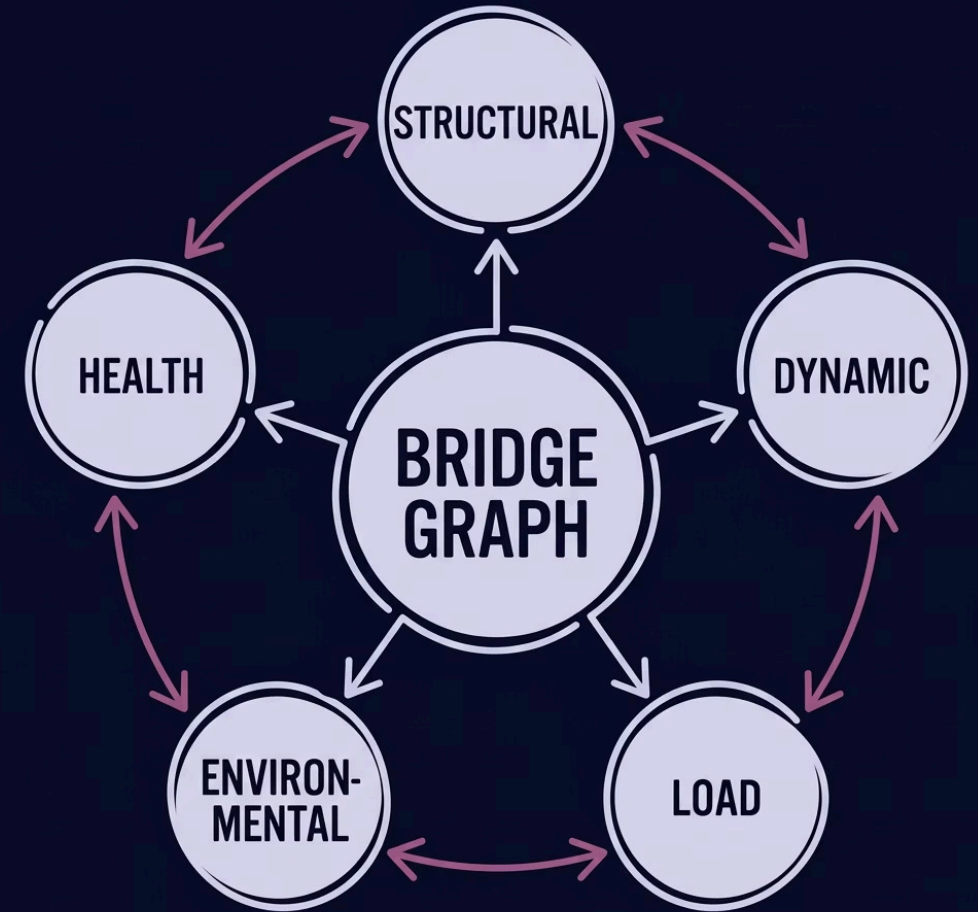
1,000 Synthetic

Balanced programmatic samples

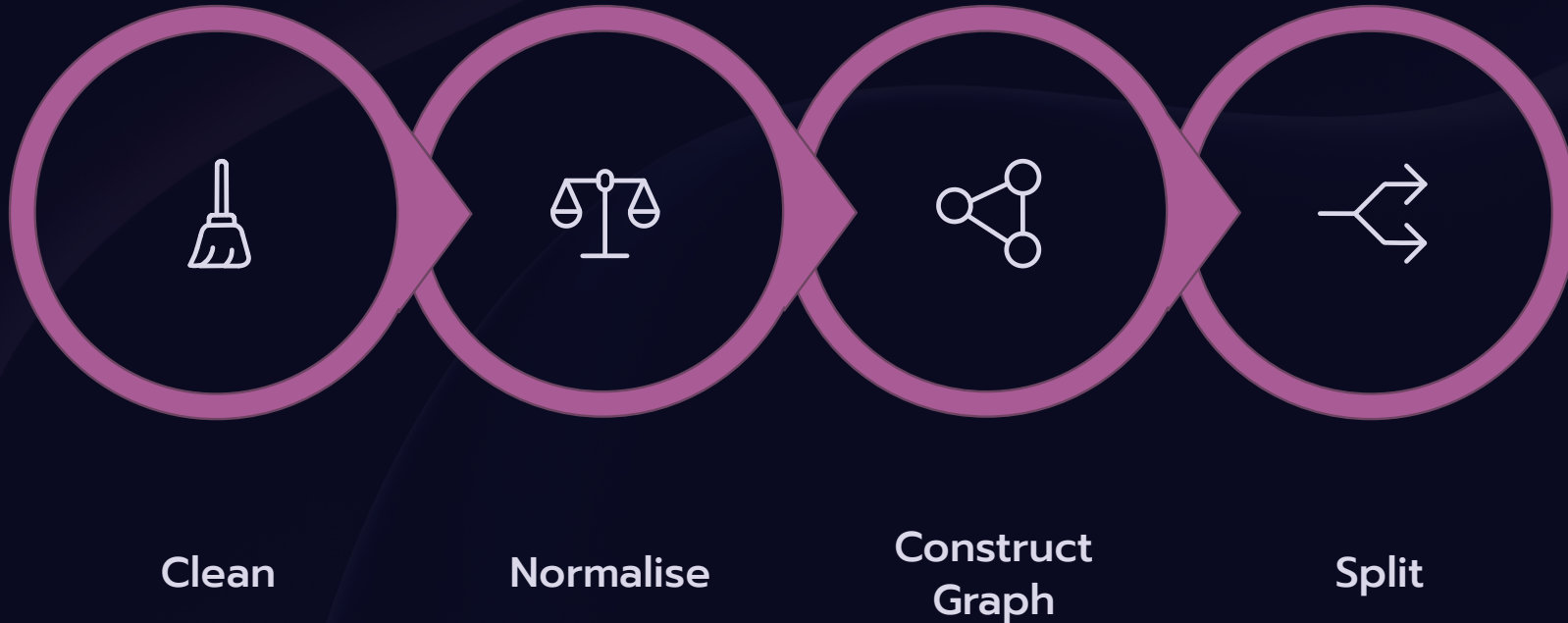
Why Graph Convolutional Networks?

Bridge infrastructure is inherently relational — strain influences fatigue, load affects vibration, environment accelerates corrosion. GCNs encode this structure directly into the model.

- Edges are defined by **physical causality** between sensor domains, not geometric proximity — a more interpretable inductive bias for structural systems.



Data Preprocessing & Graph Construction



Leakage columns (SHI_Predicted_24h_Ahead, 7d_Ahead, 30d_Ahead) are dropped before training. Each sample becomes a `torch_geometric.data.Data` object with node matrix $\mathbf{X} \in \mathbb{R}^{5 \times 5}$ and `edge_index` encoding six bidirectional physical relationships between domains.

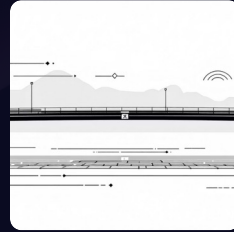
Five Sensor Domains

Each domain forms a node in the bridge graph, with edges encoding physical causal dependencies.



Structural

Strain, deflection, crack propagation, corrosion, displacement



Dynamic

Vibration, tilt, modal frequency, seismic activity, bearing forces



Load

Vehicle load, traffic volume, axle counts, impact events



Environmental

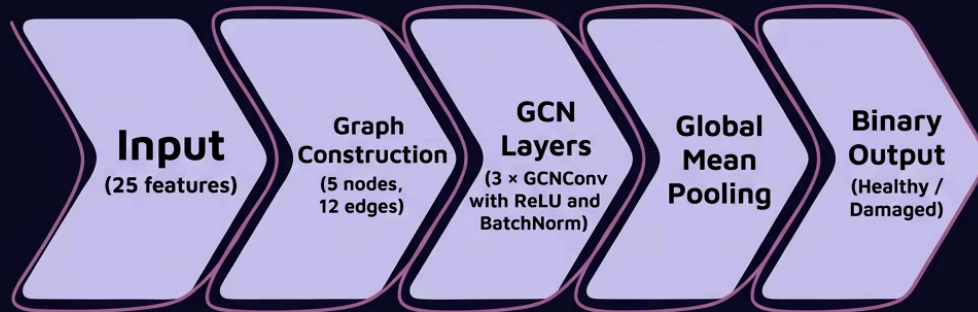
Temperature, humidity, wind speed, precipitation, water level



Health

SHI, fatigue accumulation, anomaly score, acoustic emissions

GCN Architecture & Message Passing

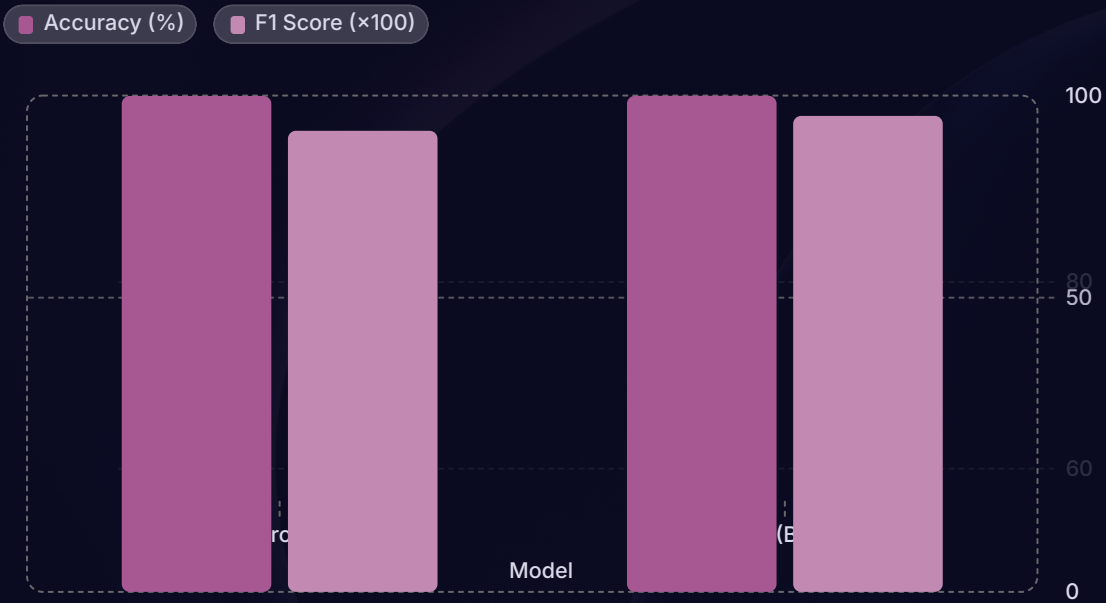


Three GCNConv layers apply the normalised adjacency update:

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)})$$

Global mean pooling collapses five node embeddings into a single graph-level vector, passed through two fully connected layers with dropout. **Total trainable parameters: 8,802.**

Results: Classification Performance

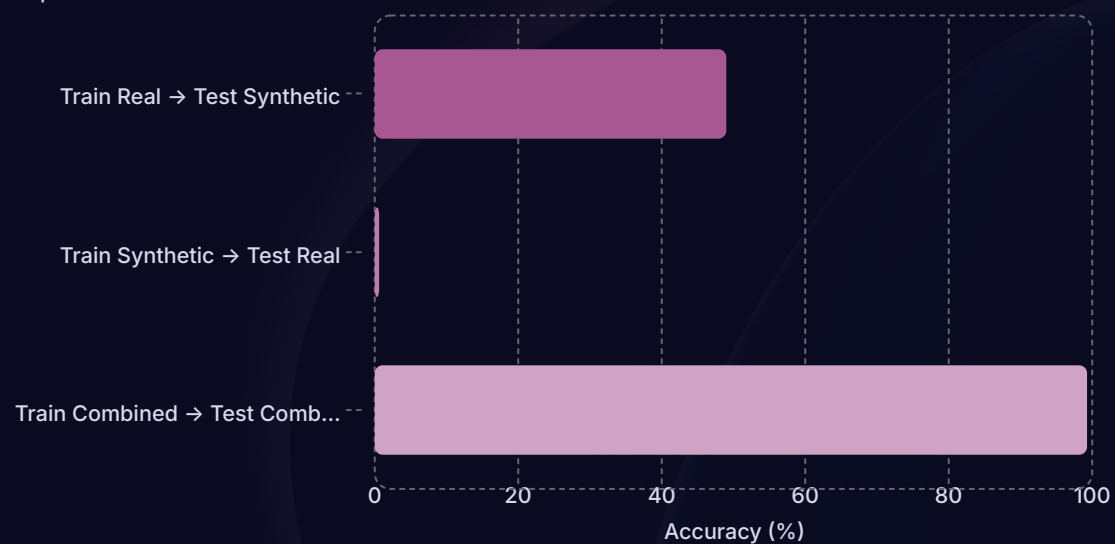


Key Findings

- Both models exceed 99.9% accuracy on the held-out 20% test split
- CNN achieves perfect precision (1.000) but lower recall — missing ~8% of damage events
- GCN's higher recall is preferable in SHM: missed detections carry greater consequence than false alarms
- GCN achieves comparable performance with only 8,802 parameters

Cross-Dataset Generalisation

Experiment



Domain Shift Analysis

- **Real → Synthetic (49%):** Near-random performance confirms meaningful domain shift
- **Synthetic → Real (0.61%):** Severe shift — synthetic data lacks higher-order sensor correlations
- **Combined Training (99.28%):** Joint training resolves domain shift; synthetic data is effective as augmentation, not a standalone source

Conclusion & Future Directions

What We Achieved

GCN achieves 99.92% accuracy on real digital twin data with causal domain-level graph topology — a more interpretable inductive bias than geometric proximity approaches.

Practical Extensions

Threshold-based explainability maps out-of-range sensor readings to ranked maintenance recommendations. A detection mechanism overrides predictions on anomalous inputs.

Future Work

Edge-weighted graphs parameterised by cable tension, dynamic topology updates as sensors fail, and transfer learning across bridge types to reduce domain shift.

